Names: Annika Gurnani, Nicholas Slazinski

Dataset: Penguins – https://www.kaggle.com/datasets/larsen0966/penguins

Calculation Functions:

calculator_1()

- For each island, how many male and female penguins have a flipper length greater than 210 mm?
    - island
    - sex
    - Flipper_length_mm

      {island: {male: num, female:num}

calculator_2()

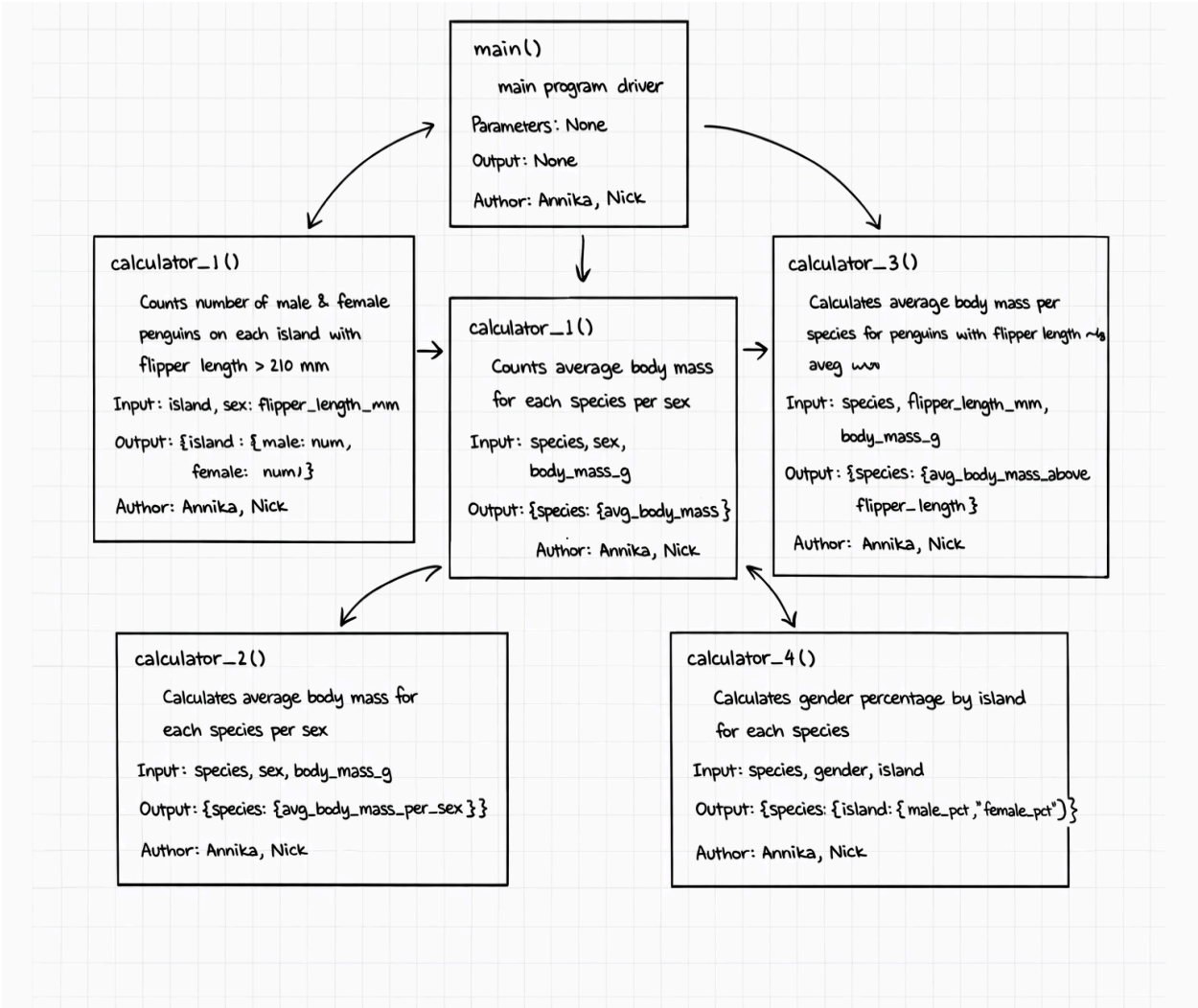
- For each species, what's the average body mass of penguins per sex?
    - species
    - sex
    - Body_mass_g

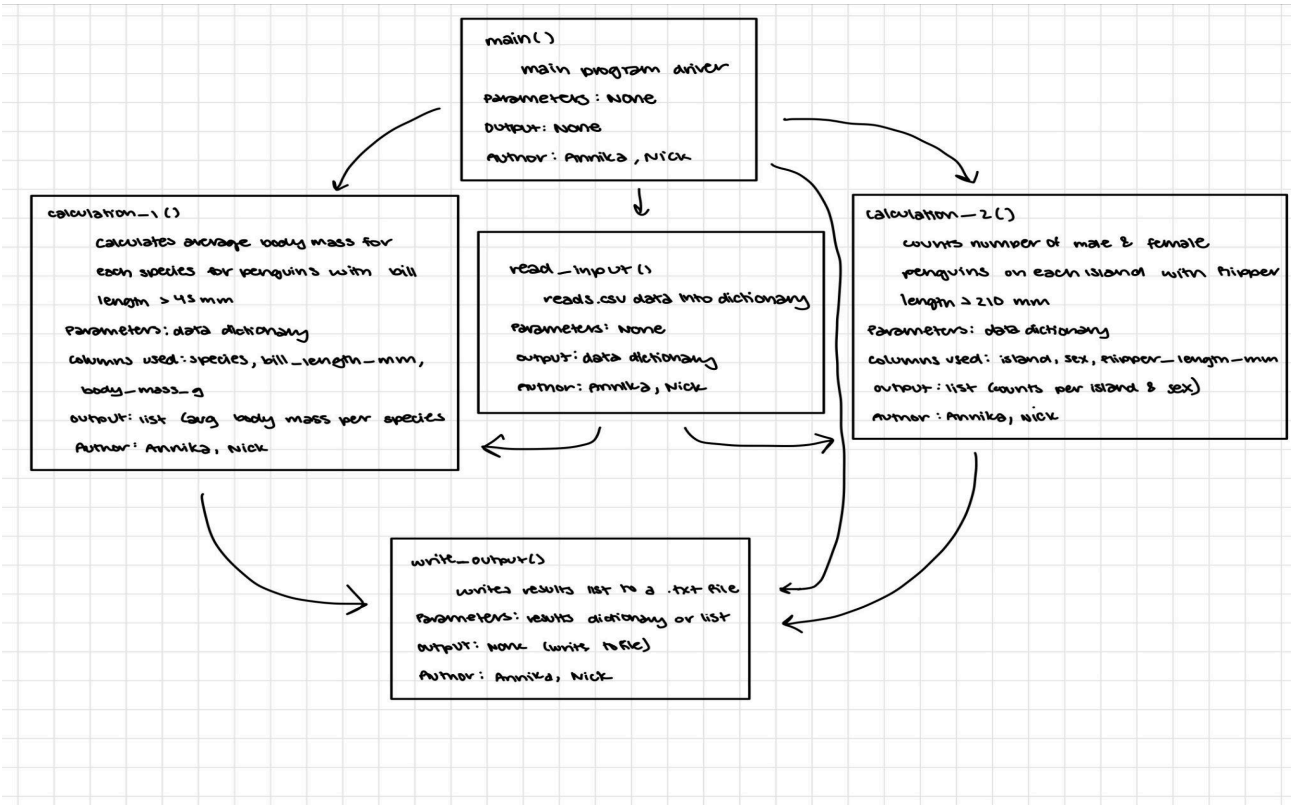
calculator_3()

- For each species, what is the average body mass above a certain flipper average length
    - species
    - flipper length
    - Body mass

calculator_4()

- For each species, what is the gender percentage by island
    - species
    - gender
    - island


main()
main program driver
Parameters: None
Output: None
Author: Annika, Nick
calculator_1()
Counts number of male & female penguins on each island with flipper length > 210 mm
Input: island, sex: flipper_length_mm
Output: {island : {male: num, female: num}}
Author: Annika, Nick
calculator_1()
Counts average body mass for each species per sex
Input: species, sex, body_mass_g
Output: {species: {avg_body_mass}}
Author: Annika, Nick
calculator_3()
Calculates average body mass per species for penguins with flipper length > avg
Input: species, flipper_length_mm, body_mass_g
Output: {species: {avg_body_mass_above flipper_length}}
Author: Annika, Nick
calculator_2()
Calculates average body mass for each species per sex
Input: species, sex, body_mass_g
Output: {species: {avg_body_mass_per_sex}}
Author: Annika, Nick
calculator_4()
Calculates gender percentage by island for each species
Input: species, gender, island
Output: {species: {island: {male_pct, "female_pct"}}}
Author: Annika, Nick


main()
main program driver
parameters: None
Output: None
Author: Annika, Nick
calculation_1()
calculates average body mass for each species for penguins with bill length > 45 mm
Parameters: data dictionary
columns used: species, bill_length_mm, body_mass_g
output: list (avg body mass per species
Author: Annika, Nick
read_input()
reads.csv data into dictionary
Parameters: None
output: data dictionary
Author: Annika, Nick
calculation_2()
counts number of male & female penguins on each island with flipper length > 210 mm
Parameters: data dictionary
columns used: island, sex, flipper_length_mm
output: list (counts per island & sex)
Author: Annika, Nick
write_output()
writes results list to a .txt file
Parameters: results dictionary or list
output: None (write to file)
Author: Annika, Nick